Consider binary classification, with $\mathcal{Y} = \{0, 1\}$, feature space $\mathcal{X}$, and hypothesis space $\mathcal{H} = \{\pi : \mathcal{X} \to [0, 1]\}$.

Exercise 1: Risk Minimizers for the Log-Loss

The first loss function of interest is the log-loss:

$$L\left(y, \pi(\mathbf{x})\right) = -y \log\left(\pi(\mathbf{x})\right) - (1 - y) \log\left(1 - \pi(\mathbf{x})\right)$$

(a) What is π_c^* (the optimal constant model in terms of the theoretical risk)?

(b) What is its risk?

(c) What is $\hat{\theta}$ (the optimal constant *score* model in terms of the *empirical* risk)?

Exercise 2: Risk Minimizers for the Brier Score

Now consider the Brier score:

$$L\left(y, \pi(\mathbf{x})\right) = \left(\pi(\mathbf{x}) - y\right)^2$$

(a) What is π_c^* (the optimal constant model in terms of the theoretical risk)?

(b) What is its risk?

Loss	Risk minimizer	Bayes risk	Optimal constant model	Risk of optimal constant model
L2				
0/1				
Log				
Brier				